

Chapter-4

The Topography and the Climate of Bangladesh

Bangladesh is one of the largest deltas of the world. The whole of Bangladesh is formed with generic hilly region, limited high land and a vast area of plain land washed by river-water. The land configuration of this country is of low and plain. The climate of Bangladesh is on the whole equable. The monsoon has a great influence over the climate. As a result, different seasons come in rotation at different periods of the year. Sometimes we feel hot and sometimes cold because of the climatic change in different seasons. A heavy rainfall takes place here due to the climate. For this reason, an excessive rainfall, unexpected flood and tidal wave influence people's lives. The people have to confront with different disasters and difficulties. In this chapter, we will be acquainted with the configuration of land, climate and natural disasters of Bangladesh.



After reading this chapter, we will be able to —

- describe the geographical location and periphery of Bangladesh;
- identify the geographical location in the World -map and the topography of Bangladesh;
- explain the classification and formation of land of Bangladesh;
- explain how the formation of land configuration extends its influence over spreading of population (habitation);
- explain the influence of spreading habitation on the use of land in Bangladesh;
- be awakened in preservation of environment;
- compare features of climate of Bangladesh to that of some countries of South Asia (like India, Myanmar and Nepal);
- explain the influence of climate over the people's lives and their livelihood in Bangladesh, and prepare a statement in this respect;
- explain the surmise of earthquake and its reasons;
- narrate situations of some countries termed as earthquake-prone regions;
- explain why Bangladesh is called earthquake prone region;
- explain preparedness and necessary steps of Bangladesh in confronting the risk of earthquake;
- be alert on encountering difficulties due to change of climate, and have capability for adaptation;
- grow supportive attitude in confronting situation after-earthquake.

Section 4.1: The Topography of Bangladesh

Bangladesh possesses a moist region comprised of alluvium. The whole of Bangladesh except some generic hilly areas at its northeast and southeast parts and some limited highlands at the northwest thereof comprises of a vast level ground washed by river water. Bangladesh is situated at the basin of three largest rivers the Ganga, the Brahmaputra and the Meghna of South Asia. The configuration of land of this country is of low and plain in nature.

The Geographical Location and Periphery

Bangladesh is situated to the South of Asian Continent in between $20^{\circ}34'$ North Axis and $26^{\circ}38'$ as well as $88^{\circ}01'$ East Meridian of Longitude and $92^{\circ}41'$ East Longitude, and the Tropic of Cancer passes through ($23^{\circ}5'$) over the middle of Bangladesh. The maximum extent of East-West 440 km, and the maximum extent from north-northwest to south-southeast 760 km. West Bengal of India, Meghalaya and Assam are situated to the North of Bangladesh; Assam, Tripura, Mizoram and Myanmar to the East; the Bay of Bengal to the South and West Bengal of India to the West of Bangladesh. A total area of Bangladesh is 1,47,570 square kilometers or 56,977 square miles.



Figure 4.1: Bangladesh in the World's Map

The Classification and the Formation of Topography

Bangladesh is one of the largest deltas of the World. A number of rivers flow through it. Among them, the main ones are the Padma, the Jamuna, the Meghna, the Brahmaputra, the Shitalaksha, and the Karnaphuli. Since the region of Bangladesh possesses a gradual sloped ground from the North towards the South, these rivers, tributaries and effluents flow from the North to the South towards the Bay of Bengal. Almost the whole region except the hilly areas at northeast and southeast side of Bangladesh, is level in shape and washed by river-alluvium.

The Entire Bangladesh possesses almost a vast area of level land. Bangladesh possesses a small area of high land. She, on the basis of her land configuration, may be chiefly divisible into three categories - the hills of Tertiary age, Flight of steps of Pleistocene age and Flood-level land of the recent age.

The Hills of the Tertiary Age: The Hills of the Tertiary Age possess an approximate 12% area of the total volume of Bangladesh. These hills were formed while the Himalayas was formed in Tertiary Age at about two million years ago. The hills existing in this part, have been divided into two categories, such as **The Hills of the South-Eastern region** and **The Hills of the North and the North-Eastern region**.

The Hills of the South-Eastern Region: This region comprises of Rangamati, Bandarban, Khagrachhori, Cox's Bazar and the Eastern region of Chittagong district. The hills and mountains existing in this part possess an average height of 610 meters. At present, the name of the highest peak of Bangladesh is Tajiodong (Bijoy) possessing a height of 1,231 meters. It is situated within Bandarban District. The second highest peak of Bangladesh is Cucradong possessing a height of 1,230 meters only. Besides, there are two other mountain peaks, Modokmual (1,000 meters) and Pyramid (915 Meters). The hills and mountains in this region comprise of sandstone, clay and shell-stone as well.

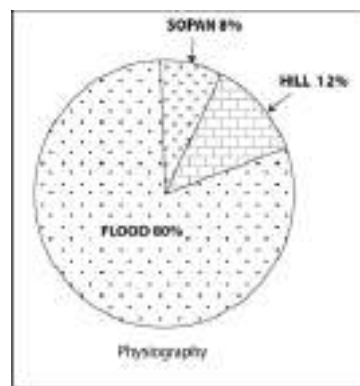


Figure 4.2 : Extent percentage of the Configuration of Land

The Hills of the North and the North-Eastern Region: This part of the Configuration of Land comprises of the hills of the northeastern region existing at the northern region of Mymensingh and Netrokona, at the north and northwestern region of Sylhet, and at the south of Moulvibazar and Habiganj. These hills possess an average height not exceeding 244 meters. The hills of the North are locally known as Tila (hillock). Their heights vary 30 to 90 meters. Chiknagul, Khasia and Jayantia are the main hills in this part.

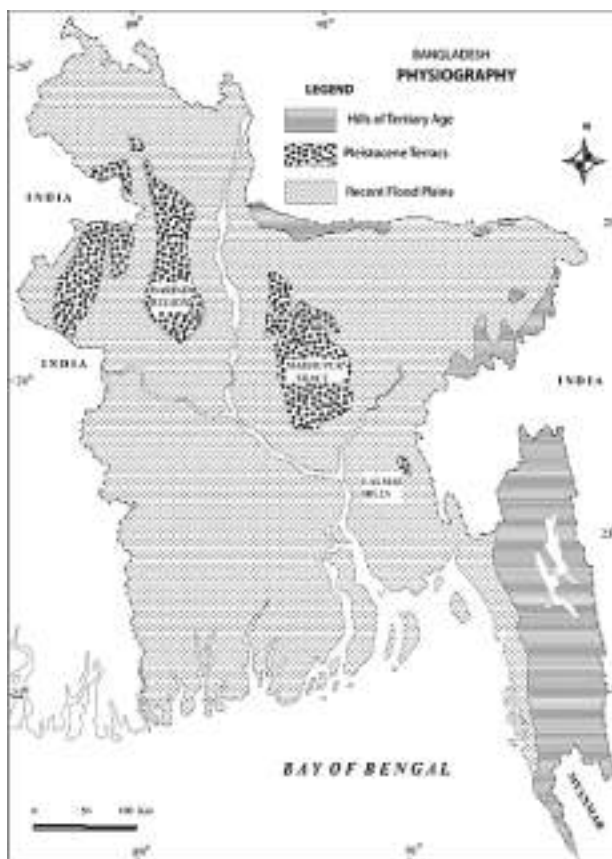
The Flights of steps/the plains of the Pleistocene age: This part comprises of about 8% of the total volume of land of Bangladesh. Approximately, the Pleistocene age is the age before 25,000 years. It is guessed that such flights of steps were formed due to originating of floods with the internally melted icy waters of that period. The flights of steps of the Pleistocene age may be divided into three categories - **The Barendra Bhumi (ancient gauda), the Modhupur and Bhawaler Gar (fort) and the Lalmai Hill**. The Barendra Bhumi comprises of

some parts of the districts of Naogaon, Rajshahi, Bogra, Joypurhat, Rangpur and a particular region of the district of Dinajpur. This region has an area of 9,320 square kilometers. The height of such areas is more than the flood level land by 6-12 meters. Soils of these areas are of grey and red colors. The courtyards of the Modhupur and Bhawal comprise of some parts of the districts of Mymensingh, Tangail, and Gazipur. It possesses a volume near about of 4,103 square kilometers. The height of such areas is more than the level land by 6-30 meters. Soils here possess radish and grey hues. The Lalmai Hill stretched from Lalmai, 8 km far away to the West from Cumilla city up to Moynamoti. It possesses an area of 34 square kilometers. The average height of these hills is 21 meters. The colour of such soils is reddish and are mixed with pebbles, sands and gravels.

The Flood Level Land of the Recent Age:

Bangladesh possesses a vast level land comprising about 80% of the total volume washed by river-water. As a number of rivers flow over the level land, they cause flood during the rainy season. Silts carried with flood water have been amassed for years after years, and formed this flood level land. This flood level land possesses a volume of 124,266 square kilometers. The soil level of the whole land is very deep and the land is very fertile. The flood level land of the recent age may be categorically divided into several divisions such as the region that spread over most parts of Rangpur and Dinajpur situated in the northwest region of the country. In fact, this region is formed with alluvium carried from the Himalayas. This region

stretches out over the districts of Dhaka, Tangail, Mymensingh, Jamalpur, Cumilla, Noakhali, Sylhet, Pabna and some parts of Rajshahi. This region comprises of Sylhet, Sunamganj, Moulvibazar, the most part of Habiganj and a small part of the district of Kishoreganj and Netrokona to the East. In addition,



4.3 : Land Configuration of Bangladesh

this level land is streched in the major parts of Chandpur and Brahminbaria and some parts of Laxmipur and Feni. This region is comprised of the alluvium carried down from the Himalayas. The Delta plains comprise of Faridpur, Kushtia, Jessore, Khulna, Barisal, Patuakhali and a part of Dhaka. The coastal level land comprises of Noakhali and from lower part of Feni river up to Cox's Bazar. The watercourse level land comprises of Khulna and Patuakhali regions and a part of the district of Barguna. The soil of these regions of Bangladesh is very fertile and plays a vital role in producing agricultural produces.

The Formation of Topography in Bangladesh and Human Habitation

Bangladesh stands 8th considering its population in the world. The population density of this country is, in comparison to its land, very excessive. Besides, the growth rate of population here is also very high. At present, population of Bangladesh is about 16.98 crore, and the growth rate is 1.12 and the density of population per km is 1119 heads (Source: Census: 2022). But, disparity of population density varies in consideration of diverse regions of this country due to formation of land-configuration.

Since Bangladesh possesses continual land configuration, people live almost everywhere on its territory. As the provision of livelihood is toilsome in the Hill tracts and the Sundarban region, density of population is comparatively very low in these two areas. Provisions of livelihood are difficult due to lack of good road and railway communications in these areas. These areas are sparsely populated due to undeveloped communication system, woodland and its land configuration. Centering round the tea industries in Sylhet, habitation has been developed here.

Individual work: Identify the land configuration of an area situated beside your school.

The level areas at the basin of rivers are formed of fertile silts. Cultivation of land is somewhat easy in such areas. That is why dense habitation has been noticed to grow in that region. Navigability of the rivers, convenience and opportunity of highways and railways attract the people to a great extent. Moreover, distribution of mass habitation is regulated due to the influence of climate over there. People prefer residing at such places having the equable condition to places having extreme climate. Weather condition in the whole area of Bangladesh is almost same, but the difference of cold and hot in the weather of northern region is relatively observed more. The mass people are preferably inclined to dwell in the level land due to the useful climate for cultivation and helpful condition for production of corns.

In Bangladesh, the regions where mineral resources have been discovered, a large number of labourers and employees gather there in quest of livelihood, and made the places crowded with habitations. Pursuing the mineral resources, agricultural resources, forest-grown assets and livestock, the mass habitation has been developed. At such places, a lot of sundry incidental industries along with the leading industries have been established. Such areas have been turned into densely populated areas based on the leading industries because of augmentation of industrial trading. Thick cluster of habitation has been developed in these districts since the industrial cities have been grown particularly in Tejgaon, Tongi, Narsingdi, Khulna, Chittagong, etc. Easy communication may be maintained if there exists enough scope for modern communication in an area either by highway, railway or by rivers. Consequently, the area becomes numerously inhabited in the course of time.

In recent days, education and culture have influenced the human society to a great extent. The region that provides much opportunity to exercise and practice education and culture, etc. the mass habitation also gets crowded normally.

The Influence of Spreading Habitation on the Use of Land in Bangladesh

Bangladesh is a densely populated country. Its population is very large comparing to its volume. Mass habitations also increased due to the growth of population. This country possesses less cultivable land comparing to its necessity. Rather its measure is getting lessened due to spreading out of further habitations which results in causing a fatal pressure over the land. Since our population is gradually increasing, a further pressure is being made on the agricultural land. The pressure increases on the land suitable for making habitations too. Already habitation problems in Bangladesh have been detected

both in villages and cities. On the other hand, a cultivable piece of land is being divided into parts among the heirs. Continuously land becomes separated into pieces, and scientific

Individual work: Identify the influences of spreading of mass habitation over land management of Bangladesh.

cultivation is not possibly made in such divided pieces of land. The farming of scientific cultivation is not possibly made in such divided pieces of land. The cultivation in scientific system is not possible due to division and distribution of the land. For rapid population growth, a lot of dwellings are being constructed on a large extent minimizing the agrarian land. A family that possessed 100 Bighas of land 30 years ago, now has 10 Bighas or less than that. In 1974, the land per head was .28 acre, which has been .25 at present. It will be reduced in near future. People have brought changes in the nature and usage of land for the excessive population. For example, people need to make their dwelling places by filling

canals, marshes with earth and cutting down trees at random. If such acts continue, natural disorder may be inevitable in future.

Section 4.2: The Climate and Natural Disaster of Bangladesh

the climate implies the daily average state of climate's constituent pervading in a vast region. The influence of climate over human lives is multifarious and important. Climate somewhat gets varied at different times over the year due to monsoon. While in the Rainy Season, a heavy rainfall takes place and vast areas of Bangladesh get overflowed. It causes excessive rainfall and untimely flood. The change of climate causes different types of natural calamities.

The Climate of Bangladesh and of Some Countries of South Asia

The climate of Bangladesh: The climate of Bangladesh is moderately warm, humid and equable. The influence of the monsoon in this region is so high that this climate is known as **tropical monsoon climate** as a whole.

Three seasons of diverse characteristics are seen in Bangladesh which harmonize the overall climate of South Asia. Those are the winter, the summer and the rainy season. Disparity in climate is somehow observed in consideration of season, but no extreme weather is perceived here like that either of a cold country or of that of the tropical country. In fine, the dry and comfortable winter season and the hot and moist summer are the main features of the climate in Bangladesh.

Features of climate on the three separate seasons prevail all over Bangladesh are described as follows:

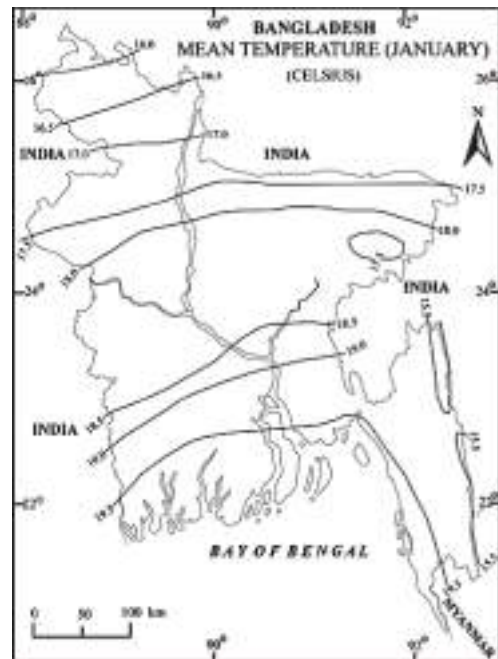
The Winter: The winter extends in Bangladesh from November to February every year. Since the Sun stays in the south hemisphere, its rays fall obliquely towards Bangladesh and in that time temperature of heat gets reduced in abundance. The highest and the lowest temperature during the winter range from 29°C to 11°C. January is the coldest month in Bangladesh. This month possesses an average temperature of 17.7°C. During this period, the temperature gradually reduces from the seashore at the South towards North, and equal-hot-lines become largely erect, and stay at the east-west side. In January, an average temperature remains 20°C in Chittagong, 19.4°C in Noakhali, 18.3°C in Dhaka, 17.7°C in Bogra and 16.6°C in Dinajpur. But, sometimes the temperature in the northern zone gets more reduced.

The Summer: The months from March to May make the Summer Season in Bangladesh. It is the hottest season of the year. The maximum temperature in this season remains at 38°C and the minimum temperature at 21°C . April is the hottest month on an average temperature. The temperature during this season gradually increases from the southern part towards the northern part due to maritime climate. Such as, an average temperature during April remains 27.64°C in Cox's Bazar, 28.66°C in Narayanganj and 30°C in Rajshahi. As the Sun draws near the tropic of cancer during the summer time in the north hemisphere, air pressure gets varied, and the southeast monsoon usually blows over the territory of Bangladesh. At that time, dry and cold air also blows from the West and northwestern sides. It is called Kalbaishakhi (North Westerlies). Besides, the coastal areas of Bangladesh are frequently hit by the devastating cyclones in April and May due to downward pressure created in the Bay of Bengal. On 29 April 1991, a cyclone along with tidal wave hit the coastal areas of Bangladesh, particularly Chittagong coastal zone, and it resulted in a heavy destruction of assets and loss of lives.

The Rainy Season: The months from June to October cover the rainy season in Bangladesh. This season starts with the arrival of monsoon at the end of June. During this period, the Sun gives its rays lengthways towards Bangladesh. That is why an excessive heat is observed here. But a heavy rainfall takes place during this period which results in increasing temperature not so high, as it was expected for the same; rather the weather always remains hot. An average temperature during this period remains approximate 27° degree celsius. June and September are the hottest months during the rainy season.

Monsoon Rainfall in Bangladesh

A Four-fifths of the total rainfall takes place during the Rainy Season in Bangladesh. An average rainfall is minimum 119 cm and maximum 340 cm. Comparatively more rainfall happens in the eastern part of the country than in the western part. For example, the amount of rainfall reaches 114 cm in Pabna, 120



cm in Dhaka, 140 cm in Cumilla, 180 cm in Srimongal and 190 cm in Rangamati. The Monsoon causes heavy rainfall in Bangladesh. During this season, rainfall takes place not less than 200 cm anywhere in the country either at the hill tracts or on the coastal belt. It pours up to 340 cm in mountainous areas of Sylhet, 200 cm in Patuakhali, 250 cm in Chittagong, 280 cm in Rangamati and 320 cm in Cox's Bazar.

The Climate of India

India possesses a vast volume of land, and diversified climates. It is because a difference happens in her axis, oceanic distance, stream of air etc. India is situated in the monsoon region. Consequently, her temperature, rainfall, moisture, etc.

entirely depend on the flow of monsoon. The climate of India may be divided into four seasons in consideration of disparity in her temperature and rainfall - the Winter, the Summer, the Rainy Season, the Autumn and the Late Autumn.

The Winter: As the Sun stays in the south hemisphere during the winter season, temperature all over India gets reduced to a great extent. The Winter covers from December to February in India. During this period, the eastern monsoon flows over India because it flows out from the frigid zone and blows through her land-surface and consequently causes blowing of dry and cold air over there. Since the Himalayas stand like a wall spreading over the northern region, dry and cold air cannot enter into India's atmosphere. Thus India gets protection from winter effect. During winter, the climate all over India remains dry, cold and comfortable. Sky remains lucid, fair and cloudless, and air possesses lesser amount of aqueous vapour.

The Summer: The months from March to May make the summer in India. The sun stays at the Equator from the south hemisphere and it gradually moves forward towards the Tropic of cancer. Since the sun begins its summer solistice, the

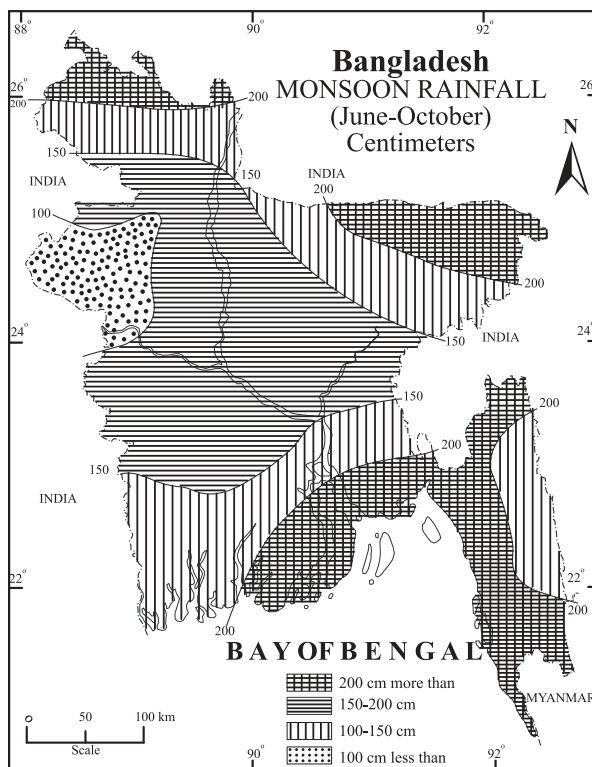


Figure: 4.5: Monsoon Rainfall in Bangladesh

temperature of India gradually increases accordingly. During this time, an average temperature of 27°C is observed at the valleys of the Ganges. Temperature gradually continues to increase as much as ahead towards the northern region. And, in that time, temperature in the desert areas at the northwest of India reaches to 48°C . Although the maximum temperature in May reaches to 43°C in Kolkata city, but an average temperature during summer does not exceed 20°C so far.



Figure: 4.6: Winter-Equal-Hot Line of India

The Rainy Season: The Rainy Season in India lasts from the months of June to September. As the Sun stays above the Tropic of Cancer at the end of June (June 21), the northern part of India serves excessively increased hot temperature (above 32°C), and temperature in the southern part there of gradually gets reduced

even below 27°C . Excessive hot temperature causes growing of a strong low pressure belt in Punjab within the northern India. The southeastern trade wind from the high-pressure belt of the Capricorn within the south hemisphere does not enter the equatorial low-pressure-belt, but moves directly towards Punjab after being attracted by the more powerful low pressure of Punjab. As this wind passes through a long way over the sea, it contains much aqueous vapour with it. That is why it causes a heavy rainfall at different regions of India after being resisted at the Himalayas and the walls of other high mountains. For this reason, an approximate 75% of the total rainfall of India pours down during this season.

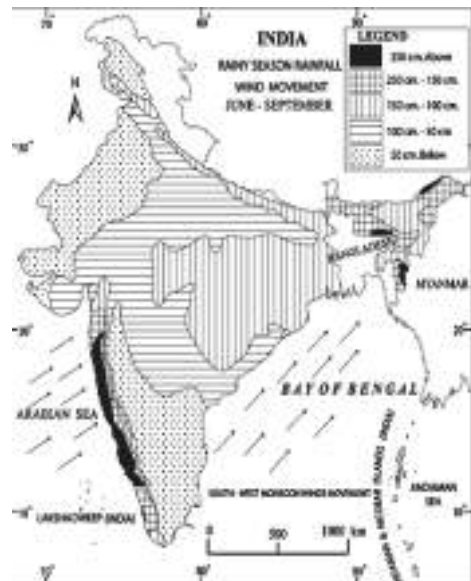


Figure : 4.7 : Rainfall and Stream of Air during the Rainy Season in India

The southwest monsoon being divided into two wings enters India, such as, the Arab oceanic wing flows over the Arab Sea and the Bay of Bengal wing flows over the Bay of Bengal.

The Autumn and the late Autumn:

October and November are the two months that make the Autumn and the late Autumn in India. During this season, the southwestern monsoon changes its direction and turns into the northeastern monsoon. It causes rainfall along with strong storm over some parts of India. Rainfall takes place during these seasons at West Bengal, Tamilnadu, Orissa and Madras coastal area. This strong storm, in the West Bengal, is called *Ashwina Jhar*. The climate and the amount of rain all over India gradually reduces at the end of the Dewy season.

The Climate of Myanmar

The Climate of Myanmar is similar to that of the Indian Subcontinent and seems like tropical monsoon. The climate of this region possesses a clear presence of three diverse seasons in consideration of differences in her temperature and rainfall, namely the Winter, the Summer and the Rainy season. The season wise features of climate of this region are discussed as follows:

The Summer: The Summer in Myanmar comprises of the months from March to May. During this season, most part of the country gets very hot and an average temperature reaches almost to 29°C. The Sun stays in the north hemisphere during this season and a great low pressure is formed in the middle of Asia. Monsoon begins to flow over there. During this season, temperature remains 19°C in Bhamo, 32°C in Mandalaya and 27°C in Yangoon as well.

The Rainy Season: The Rainy season in Myanmar comprises of the months from June to October. A heavy rainfall takes place in this region due to the southwest monsoon, and at the middle of May, normal rainfall takes place in Yangoon, and at the end of that month, it prevails all over the country and continues to do so until October. A great difference in the amount of such rainfall is observed in different regions of Myanmar. A heavy rainfall persists up to 200 cm on the coastal belt of Arakan and Tenasserim for the monsoon in the southwestern part, whereas only 80 cm rainfall takes place in the northern hilly regions.

The Winter: As the sun stays in the south hemisphere during this time, a great high pressure is formed in the middle of Asia in the north hemisphere. A low pressure is formed from here in the relatively more hot areas in comparison to that in the sea at the southeast side that causes the blowing of the northeast trade wind towards the southwest side instead. Although it is expected that Myanmar would

possess excessive cold due to the influence of cold stream of air in the North, but cold cannot be so severe due to the existence of hilly region in the North. This stream of air prolongs to blow till the month of March. During this time, snowfall pours over the high hilly region in the North of Myanmar, and its temperature reaches almost at the freezing point.

The Climate of Nepal

In Nepal, considering its existing temperature and rainfall in the climate, two seasons are clearly perceived over there. Heavy rainfall persists in this region during the months from

Group work: Make a list showing features of the climate existing in Bangladesh, India, Myanmar and Nepal.

June to September, and this period is called the rainy season. In July temperature at Katmandu remains 24.4 °C. On the other hand, a very dry condition and lack of rains persist over there from the months of November till January in Nepal. The temperature is also reduced during this season to a great extent, so, it is considerably termed as the winter. The temperature at Kathmandu remains approximately 10°C in January. The condition of temperature of any other parts of Nepal does not increase at the high hilly regions, and much difference of temperature in between cold and hot is observed over there. The yearly average rainfall in Nepal is 145 cm, most of which takes place within the months of June to September.

The Influence of Climate on the People's Life and livelihood

The Influence of Climate upon the People's Life and Livelihoods in Bangladesh
The socio-economic condition of the people of Bangladesh depends largely on nature and natural resources due to its geographical position and the climate as well. A lot of changes take place in people's life and livelihoods here due to the change of climate. Depending on its climate, frequent natural calamities and disasters, long-lasting floods, disintegration of rivers cause changes in the living and livelihoods of the people. A heavy rainfall takes place in Bangladesh due to the influence of the southwest monsoon during the rainy season. Consequently, diverse crops and fruits grow here. A little rainfall takes place in this country during the winter season due to the influence of the northeast monsoon. Consequently, diverse rabi crops like paddy, wheat, tobacco and different sorts of pulses, oilseeds, potatoes, onions, garlic, coriander, etc. and various vegetables, can be produced here, flood is caused by this wind. Silts carried by flood water overflowed to the agricultural land that enhance fertility of agricultural lands, and cause to grow better crops as well.

An average temperature across the country has increased due to the change of climate. A heavy rainfall is taking place during the rainy season and sometimes

the advancing of the rainy season is delaying. The heavy rainfall within a short time, landslide, flood and hill-slide due to heavy rainfall are the common phenomena here. Sea level has heightened. The sea level of the *Hiron point*, *Char Changa* and *Cox's Bazar* has also been increasing by 4 mm to 6 mm every year.

The rivers flowing through this country protect human lives and their livelihoods as well. Human life-styles are experiencing change because of the disappearance of the rivers. The flow of water

Individual work: Identify the influence of change of climate over the environment.

in the common rivers is decreasing due to the impact of dams and river linking projects built upstream by neighboring India. Besides, about lakhs people have become homeless because of river erosion, and have taken shelters in cities for their livelihoods. A number of small farmers are getting employed in the cities due to obstruction in agricultural production, which consequently causes disintegration of family, insecurity of children, aged persons and women as well. Living and livelihoods have a close relation to the environment they exist in. A number of creatures of the animal kingdom, being failed to cope up with the change of climate and with environment has been extinct, animal variety has been vanished and food production has been reduced in the course of time. Accordingly, the hunger and poverty have increased. The fishermen residing at the coastal belt are changing their profession and a number of fishermen are moving towards the cities for their livelihood in quest of employment over there.

Livelihoods of the coastal area people, by any means, depend on the natural resources. People residing at this region, the poor people, the hardcore poor, middle class people and the rich people pass their lives depending on natural resources like water body, *beels*, *haures* canals, arable land, gardens, trees and plants, fish, etc. The foundation of their living existence is getting lost due to different kinds of disasters because of the change of climate. Various natural calamities such as water logging, salted water, flood, deluge, storm, Lightning, *Kalbaishakhi*, cyclone, etc. have influenced human lives and their livelihoods immensely. The number of displaced people is increasing in the cities due to unplanned development and climate change.

Lightning: Lightning is a natural phenomenon. Recently, it has been observed that the number of deaths due to lightning is increasing. According to the Department of Disaster Management, an average of 350 people have died in Bangladesh annually in recent years.

What is Lightning? Lightning is a natural phenomenon that occurs during a thunderstorm, accompanied by a flash of bright light and a loud noise. The most lightning strikes occur in the northeastern districts of Bangladesh in the months of April and May.

Ways to avoid lightning-

1. No place outside the house is completely safe. Water bodies or under trees are not safe. You should stay indoors during lightning. You should not leave the house until the lightning has completely passed.
2. If lightning strikes while you are working in the field and there is no safe shelter, you should sitting curled up. Under no circumstances should you lie down on the ground.
3. If lightning strikes while you are on a boat, you should take shelter under the boat's canopy and stop fishing.
4. It is completely safe to touch a person injured by lightning. Therefore, do not panic and take the injured person to the nearest hospital immediately.

The Concept of Earthquake

Earthquake is one of the natural disasters. This tendency of earthquake is observed almost in a number of countries and regions of the world. Earthquake is liable for lots of devastation in human civilization. It is supposed that approximate 15 million people died because of earthquake over the last 4000 years. The earth-surface is being transformed through different natural processes. The scientists resolve that heat from the radioactive substances is dispersed beneath the earth's crust. This heat being amassed causes convectional stream within the Earth, and there generates a strong force which causes agitation and metamorphosis at different parts of the crust. Such metamorphosis within the crust by the internal force takes place either in sudden course or in slow course. Earthquakes and volcanoes are major force among all other forces causing metamorphosis in sudden process.

The Earthquake

Sometimes some parts of the earth-surface get an unexpectedly shake for some reasons. This quake or shake turns to severe state from its very tender state which

normally lasts for few seconds. Such sudden and momentary vibration of the surface of earth is called 'earthquake'. The region within the interior side of the earth where an earthquake is formed is called the Focus of earthquake. The name of the outer surface of earth at just above the focus is called Epicenter. The velocity of the quake from the Epicenter is gradually reduced around.

The Reasons for Earthquake: While searching the possible reasons for earthquake, the scientists noticed that some regions of the world are earthquake prone. In this zone, the new mountain ranges are situated. The cross section of the Earth shows that the crust is comprised of eight very large pieces and six regional pieces. These are known as the 'Tectonic Plates'. Of the many reasons earthquakes happen, the displacement or dislocations of these plates are the most significant reasons. In addition, the other reasons for earthquakes are as follows - earthquake takes place when any sudden shakes stimulate the earth all along the

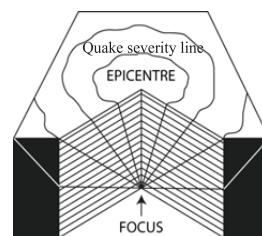


Figure 4.8 : Focus and Epicenter of earthquake

detachment or crack of the foundation rock. Besides, earthquake also takes place while magma of a volcano comes out of the interior side of the earth. When the earth's crust gets compressed by dispersing heat, crack or plait is created in order to maintain consistency of load at the rock-level down the earth, and a quake of earth is also observed. When rock is collapsed or displaced at any place of the earth's crust due to earth-agitation, a vibration in earth is observed. Besides, when either of the two plait's staying abreast enters the bottom all along the others' periphery or fluctuates horizontally, a quake in earth is observed within such fractious environment.

Individual work: Identify the reasons causing an earthquake.

The Consequences of Earthquake

A lot of changes and damages take place in the world due to earthquake. A direct or indirect consequence of earthquake is discussed below:

An innumerable cracks and dislocations are created in the earth's crust because of earthquake. Sometimes a vast area of land at the bottom of the sea rises for the earthquake.

Group work: Make a list showing the impact of the earthquake.

Again, sometimes a large area of land at the surface submerges at the bottom of the sea. Sometimes, a river-motion is changed or blocked. Big ice-pieces suddenly fall down from the hillsides because of the earthquake, and causes heavy damages of the hills. The sea level gets down from the shore for the shake of the quake, and then the sea water falls out towards the shore in the form of waves arising at a height of 15-20 meters with horrible roar. Such tidal waves are

called Tsunami. On 26 December 2004, with the blows of Tsunami created due to earthquake, innumerable lives and assets were lost in different countries such as in Indonesia, Malaysia, Sri Lanka, Thailand and India. Sometimes, the highlands get submerged into the sea-water. And sometimes, a place at the bottom of the sea gets up and forms islands in the sea. Because of the influence of earthquake, a horizontal side-pressure on the surface of the earth persists. Consequently, it gets contracted and makes fold. Landslide gets sloped from hilly region for earthquake, and resists the river-motion which causes the formation of a lake. Because of earthquake, famine and devastating epidemic continue in the damaged area for which a heavy loss of lives takes place. The telephone lines, electricity, etc. get disconnected, and make the communication system disrupted. The road communication is damaged too. If a strong earthquake takes place at the bottom of the sea, big dams, culvert and bridges existing on the surface are damaged. Sometimes, it causes Tsunami.

The Steps to Minimize the Losses of Earthquake: A correct forecast on any natural calamity may help minimize the measure of destruction. Particularly, correct forecast on the origin of disasters, duration of staying, force limit and the areas that may be victim by the same is very important. The nature of earthquake is different from that of the other disasters. Moreover, it takes place suddenly, continues momentarily, and occurs within the interior of the earth. That is why there is no scope to forecast it directly. If some steps are taken, it may help guess about an earthquake. The scientists have guessed that the regions where no earthquake has taken place during the last 100 years but are normally identified as earthquake prone zone, the possibility of earthquake is greater in that zone. Among this region, the most mentionable are South California, Middle Japan, Middle Chile, Taiwan and the west coastal areas of Sumatra, etc. A high-degree earthquake measured in the Richter scale may take place in these regions within several decades to come. It is possible to minimize the losses if the following steps are taken in the earthquake prone regions.

While constructing any infrastructure anywhere, the soil should be tested by experts, and the foundation of the structure should be made firm and strong. A safe distance should be maintained in between two constructions that must have several doors instead of one, and any multi-stored building should possess emergency exits where necessary. Lines of electricity, gas, etc. should be maintained in a faultless state. The furniture of the house should preferably be made of wood as much as possible. There should be a helmet for each of the family members in a residence. At the time of earthquake, switches of electricity and gas line should be kept off, and people should take shelter in an open space.

Bangladesh because of the collisions happened in tectonic plaits. Bangladesh stands in between the earthquakes prone Indian plate and the Myanmar sub-plate. The geologists say that liniamote extends over the region starting from Assam up to Pagla of Dhaka. This liniamote is linked with the Assam Douki Danger-fault at the East-West. Sylhet of Bangladesh is situated at this Danger fault-line. A French Engineering Consortium drew a map mixed with earthquake belts of Bangladesh in 1989. These belts have been shown therein. It termed the first belt as 'Terribly Disturbing' the second one as 'Dangerous' and the third one as 'Venial'. These belts are called 'Seismic Risk Zone'. Bandarban, Chattogram, Sylhet, Mymensingh and Rangpur are situated in the first belt; and Dhaka, Tangail, Bogura, Dinajpur, Cumilla and Rangamati in the second belt; and all other regions in the third one as well.

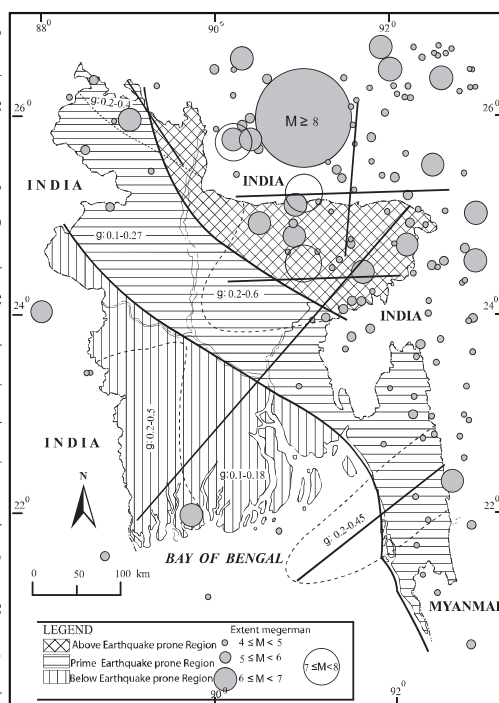


Figure 4.10: Earthquakes Prone Region of Bangladesh

The Preparations and Necessary Steps of Bangladesh to Confront the Risk of Earthquake

No minimum preparation exists in Bangladesh to confront topsy-turvy of any severe earthquake. For its poor rescuing measures and capability, a medium scale earthquake may cause a catastrophe in Dhaka. Millions of people may be the prey of death in a helpless situation. The risk of earthquake is getting worse as the population density, unplanned cities, lots of multi-storeyed buildings, lack of open spaces, narrow lanes, poor rescuing measures and equipments. If all the concerned people with earthquake can consciously take necessary steps, it may be possible to restrain the earthquake and to minimize its losses to a great extent. If we take a careful look, we will notice that an earthquake takes place in our country for 30-35 seconds, more loss of lives will occur because of the delay and failure in rescue operation than the loss of lives may be caused by the same. For this reason, it is necessary for us to have proper idea on preparation in confronting the earthquake. An earthquake may cause a heavy loss within short time, and lasts for short while. It suddenly occurs within the interior side of earth. Consequently, there exists no scope to directly observe the earthquake anyhow. Notwithstanding, the geologists can refer to some advanced steps to confront the earthquake which may largely help minimize the loss.

The Preparation of Bangladesh in Confronting Risk of Earthquake

The preparations that should be taken for an earthquake are as follows:

Those who will construct new buildings should comply with the National Building Code while compiling structure and designs for the same accordingly, and they should construct their buildings with the quality construction materials under the supervision of expert engineers as well.

While constructing walls with bricks, no building should be built above 4-storey height; if a building is above two storey-heights, an erect steel rod should be put in between the bricks at each corner, and an erected steel rod should be put in between each windows and door too. It is not good to make windows or doors at corner-side any way. If such alertness is taken into consideration, brick-built buildings' resistance capacity increases. Although cracks may take place on the wall during earthquake, they will not collapse so easily. But, it will cost more 1 % to 2% than the average construction cost.

Tin-shed houses in the village have hardly in a risk because of the earthquake. But the house made of earth have no resistance power. In order to increase the resistance force, bracing of wood or bamboos may be diagonally used to serve the purpose.

If a building does not possess so resistance force, the building may further be made strong even after its construction work. In respect of concrete building, volume and shape of poor sites can be increased with concrete moldings by using extra amount of rods. Moreover, coating on walls by using Ferro cement may increase the resistance force. The ground floor should not be kept vacant. Resistance force of Semi-brick-built rooms may be increased by fastening the same around in the form of dragging.

The Dos and Don'ts of a Person in the Preparation of an Earthquake

As a part of the preparation of earthquake, one should keep a battery-run radio .and a torch light in the house. He should keep a first-aid box ready to administer.

Individual work: Make a list of the responsibilities of a person during the earthquake.

He should know where the main switches of gas, water and electricity lines of the house remain and how to turn them off. He should identify the most protected site within the house. He should keep the contact numbers of hospitals, Fire Brigade, etc. He should also provide the concept of earthquake to the children at schools, and instruct them on the do's and don'ts at the time of earthquake. Finally, he should stay away from the brick-built buildings.

The dos and don'ts of mass people during earthquake

Each and every one the common people keeps themselves easy and tranquil at the time of an earthquake. He should run away from the house if it is one-storeyed construction. He should not be greedy confining at the home for something. He should not enter the house if he stays outside at the time of earthquake. If earthquake takes place during night and if he stays within any multi-storied building, he should stay under the table or the cot and should stay away from glass-windows. If it is necessary, he should take shelter at the corner of the room or at the bottom of the column.

If anybody stays outside the house at the time of earthquake, he should stay away from any large trees, lines of electricity, water, gas, etc. He should not try to jump from any high-rise building, windows or roofs. If he is driving on the road, he should not continue driving but should keep the engine stopped. Landslide is most likely to take place at hills, mountains, high ditch or sloping land, one should take shelter at safe places from such risky sites.

The Responsibility of a Conscious Individual after Earthquake

While earthquake is over, a conscious people should examine his own injury as well as the injuries of others and provide first aid where necessary. He should check the lines and connections of water, gas, electricity, etc. He should open the windows and the doors of the house. He should keep the radio on in order to hear announcement broadcast centrally regarding confrontation of the disaster. One should not walk bare footed. He should be aware of plundering and work according to the advice of the experienced individuals.

Exercise**Multiple Choice Questions**

1. In which district is the highest mountain peak of Bangladesh located?
 - a. Chattogram
 - b. Khagrachhari
 - c. Bandarban
 - d. Rangamati.

Read the following passage and answer questions 2 and 3.

Ristana went to a fair with her parents on a December afternoon. Suddenly, it started raining, and they quickly took shelter under a tree. However, after a slight rainfall, the clouds cleared.

2. Which wind did influence the rainfall Ristana witnessed?
 - a. Northeast Monsoon
 - b. Northwest Cold Wind
 - c. Southwest Monsoon
 - d. Southwest Dry Wind

3. Which crop can be produced due to the said rainfall?

- | | |
|----------|-----------|
| a. Maize | b. Jute |
| c. Wheat | d. Cotton |

4. Due to climate change:

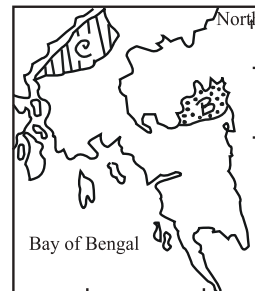
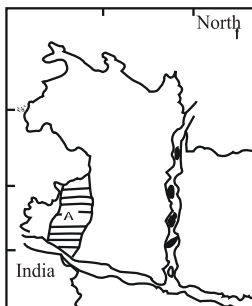
- i. Professional changes are occurring in people's lives
- ii. Sea level is rising
- iii. Various species of animals are becoming endangered

Which of the following is correct?

- | | |
|---------------|------------------|
| a. i and ii | b. i and iii |
| c. ii and iii | d. i, ii and iii |

Creative Questions:

1.



- a. What is an earthquake?
- b. Why do the rivers of Bangladesh flow towards the Bay of Bengal?
- c. Explain the physiographic structure of the 'A' marked region on the map.
- d. Is there a difference in population density between regions 'B' and 'C' in the image? Give your arguments in support of your answer.

Short Answer Questions

- 1. What are the latitude and longitude coordinates of Bangladesh?
- 2. Why the hills in the north of Bangladesh are called 'tila' (hillocks)?
- 3. What is meant by 'Ashwina storm'?
- 4. What is the epicenter of an earthquake?
- 5. What is a 'Seismic Risk Zone'?